

DIGIVOTE: DIGITAL VOTING SYSTEM

Comprehensive Platform Architecture, End-to-End Workflows, Security Protocols, and AI Knowledge Specification

Document Version: 2.0 (Official)	Classification: Public Civic Specification
Target Audience: Voters, Organizers, Auditors, AI Assistant	Primary Backend Engine: Django REST Framework + Gemini + Pinecone
Security Standard: AES-256-GCM / WebAuthn FIDO2 / ArcFace	Frontend Framework: React 19 + Vite + Tailwind CSS

Executive Summary & Core Mission

DigiVote is a high-assurance, tamper-evident digital voting platform engineered to conduct secure, transparent, and verifiable elections for civic governments, academic institutions, corporations, and community organizations. The platform balances two traditionally conflicting imperatives in digital democracy: **strict voter identity verification** and **absolute ballot anonymity**.

To guarantee integrity, DigiVote enforces a *Triple-Separation Architecture* across the entire lifecycle:

- Citizen Identity Layer (Accounts/Authentication):** Governs user accounts, passwords, MFA OTP challenges, session tokens, and security audits.
- Election Eligibility Layer (Elections):** Enforces bounded voter rosters, registration rules, candidate nomination slates, and multi-stage biometric verification.
- Cryptographic Ballot Box (Voting):** A zero-knowledge blind repository where ballots are envelope-encrypted with AES-256-GCM. Ballots store no user foreign keys, IP addresses, or traceable timestamps.

1. System Architecture & Technology Stack

DigiVote is designed as a modular decoupled system consisting of an asynchronous REST API backend, an ultra-responsive Single Page Application frontend, a relational persistent store, and a dual-stream RAG AI Assistant:

Component	Technology	Key Responsibilities
Frontend	React 19, Vite, Tailwind CSS, Lucide Icons	Client-side dashboard, accessible voting booth, live theme toggle (Light/Dark/System), AI chat UI.
Backend API	Python 3.11, Django 5.x, Django REST Framework	RESTful endpoints, business logic validation, state machines, atomic database transactions.
Database	PostgreSQL / SQLite, UUIDv4 Primary Keys	ACID transactions, row-level locking (select_for_update), relational voter rolls and candidate slates.

Authentication	SimpleJWT, HMAC OTP, WebAuthn / FIDO2	Short-lived bearer access tokens, rotating refresh tokens, session tracking, device fingerprinting.
Biometrics	ArcFace, MediaPipe, OpenCV	Webcam facial landmark liveness detection, 512-dimensional facial embedding match against voter KYC photo.
AI RAG Pipeline	Pinecone Vector DB, Google Gemini LLM	Semantic vector search (gemini-embedding-001, 3072 dims), authorized DB context synthesis, generative answers (gemini-2.5-flash).

2. User Roles & Authorization Matrix

DigiVote operates on a Principle of Least Privilege (PoLP). Every request is evaluated against user credentials and contextual ownership:

Role Name	Scope & Definition	Permitted Capabilities
Citizen Voter (VOTER)	Standard authenticated citizen.	View public & assigned elections, verify identity via OTP/facial scan, cast single anonymous ballot, download receipt, consult AI Assistant.
Election Creator (ELECTION_CREATOR)	Authorized election organizer.	Create draft elections, upload voter roll CSVs, configure candidates/photos, set verification requirements, schedule, pause/resume, and publish results.
System Administrator (ADMIN)	Platform security officer.	Global user management, institutional oversight, emergency election cancellation, system health monitoring, cryptographic audit inspection.

3. Complete End-to-End System Workflows

3.1. Workflow 1 & 2: Citizen Registration, Verification & Multi-Factor Login

Every citizen participating in DigiVote must undergo an authentic onboarding sequence. Account creation requires full name, valid national/institutional email, phone number, and a secure password. An email verification link with a cryptographic token is dispatched immediately. Once confirmed, the account transitions from PENDING_EMAIL_VERIFICATION to ACTIVE.

Multi-Factor Authentication (MFA) Protocol:

1. Citizen enters email and password into the React portal.
2. Backend verifies hashed credentials (Argon2 / PBKDF2). If valid, a PENDING_MFA pre-authorization challenge token is issued.
3. A 6-digit cryptographic OTP is generated, hashed with SHA-256, stored in OTPVerification with a 5-minute expiry, and sent via email.
4. Citizen inputs the OTP into the 6-digit input mask.
5. Backend validates the code within maximum attempt limits (3 attempts). Upon success, full JWT Access and Refresh tokens are issued.

3.2. Workflow 3: Device Session Tracking & Session Revocation

DigiVote continuously monitors active sessions to detect anomalous concurrent access or compromised credentials. Each successful authentication records the IP address, user-agent, operating system, and unique JWT token ID (jti) into UserSession. Citizens can view all active devices from the Sessions & Security dashboard and instantly revoke suspicious sessions or trigger a global *Revoke All Other Sessions* command.

3.3. Workflow 4: Election Creation, Rules & State Machine Lifecycle

Elections follow a strict, non-reversible state machine managed by the backend engine to ensure chronological fairness:

Status Code	State Description	Allowed Operations
DRAFT	Initial creation state.	Edit title, description, election type. Upload voter rosters and candidate nominations.
CONFIGURED	Setup finalized and locked.	Define verification rules (OTP, Webcam, Biometric) and results visibility schedules.
SCHEDULED	Timers armed for automated start.	Countdown timer active. Voters can preview election details and test verification.
ACTIVE (Live)	Voting polls are officially open.	Eligible voters can access the voting booth and cast ballots. Organizer can view live turnout statistics.
PAUSED	Temporary emergency freeze.	Organizer pauses voting in case of network disruptions or security investigation. No votes accepted.
COMPLETED	Polls closed; tallies computed.	Automatic official tally calculation. Results published immediately or on scheduled date.

3.4. Workflow 5 & 6: Candidate Configuration & Voter Roll Management

Candidate Slate Setup: Organizers add candidates with full legal name, political or organizational affiliation, biographical profile, and verified photo. Display order can be reordered atomically via POST `/api/elections/{id}/candidates/reorder/` or configured for randomized ballot presentation to eliminate positional bias.

Voter Roll Configuration: To vote in an election, a citizen's email must exist on the election's official EligibleVoter roll. Organizers can add individual eligible voters or execute a **Bulk CSV Upload** (e.g. university rosters, corporate directories, or municipal registries). The system links registered user accounts with eligible records while maintaining independent participation records.

3.5. Workflow 7: Multi-Stage Voter Verification Gateway

Before entering the voting booth, citizens must clear the election-specific verification gate configured by the organizer:

- **Level 1 (Email OTP):** A unique 6-digit challenge dispatched to the voter's registered email.
- **Level 2 (Webcam Liveness Detection):** The browser captures video frames analyzed via MediaPipe/OpenCV to verify 3D human presence and prevent spoofing.
- **Level 3 (Facial Biometric Match):** The citizen's live face embedding is matched against their government KYC profile photo using ArcFace (512-dimension cosine distance threshold ≥ 0.68).

Upon completion, an encrypted, time-bounded `VotingAuthorizationToken` is issued permitting access to the polling booth.

3.6. Workflow 8: Secure Polling Booth & Cryptographic Ballot Casting

The voting sequence is protected by mathematical vote-secrecy and strict database concurrency controls:

- 1. **Ballot Presentation:** The frontend renders the ballot candidates. The UI presents clean selection buttons with a confirmation modal. No voting action or automated vote-casting is permitted via third parties or AI assistants.
- 2. **Atomic Server Transaction:** When the voter clicks *Confirm & Submit Ballot*, the request enters an atomic database transaction (transaction.atomic()) with row-level locking (select_for_update) on the voter's receipt record.
- 3. **Double-Voting Collision Defense:** The system verifies whether a VoteReceipt exists for (voter_id, election_id). If a duplicate request is received concurrently, the unique database constraint collides immediately and the transaction rolls back.
- 4. **Zero-Knowledge Ballot Recording:** The anonymous ballot is encrypted with AES-256-GCM and stored in voting_ballot. Crucially, the ballot record contains **NO voter foreign key, NO user identifier, and NO IP address**.
- 5. **Cryptographic Receipt Generation:** A unique SHA-256 receipt token is computed: SHA256(voter_id + election_id + cryptographic_salt). The voter receives this receipt to independently audit that their vote was included in the election tally without exposing who they voted for.

3.7. Workflow 9, 10 & 11: Real-Time Turnout, Tamper-Evident Auditing & Official Results

Turnout Monitoring: Organizers have access to live aggregated turnout graphs (percentage of registered voters who have cast ballots) without revealing candidate vote shares during polling.

Cryptographic Audit Trail: Every critical event (election state transitions, voter additions, configuration updates, emergency pauses) is recorded in AuditLog with SHA-256 integrity hash chains ensuring retroactive tamper-evidence.

Official Tally & Reports: Once the election reaches COMPLETED status, the tallying engine executes an automated sum of all valid ballots. Certified reports detailing participation numbers, percentage distributions, candidate rankings, and cryptographic verification seals are generated and exportable as certified PDF and CSV files.

3.8. Workflow 12: DigiVote AI Assistant RAG Pipeline

DigiVote features an integrated civic AI Assistant engineered to guide voters through procedures, rules, and election logistics. The assistant employs a **Dual-Stream Context Architecture** combining document knowledge retrieval with live database context:

Step	Subsystem	Operational Mechanics
1. User Query	React Frontend	User types question in AI Assistant. Query is validated (max 2000 chars, no empty input). Dispatched via POST /api/ai/chat/.
2. Document RAG	Pinecone Vector DB	Query is embedded with gemini-embedding-001 (3072 dims). Pinecone queries 'digivote-docs' namespace to extract top-k relevant policy text chunks.
3. Live DB Context	PostgreSQL ContextService	If election_id is passed, queries read-only database state: user eligibility, election status, start/end dates, and candidate names.
4. Guardrails Synthesis	ContextBuilder	Assembles prompt with strict system instructions: prompt injection defense, strict candidate neutrality, refusal of voting persuasion.
5. Answer Generation	Gemini 2.5 Flash	Synthesizes accurate civic answer with page citations. Does not recommend candidates or execute voting actions.

4. Comprehensive REST API Endpoints Specification

All API endpoints require HTTPS and standard JSON payloads. Authenticated endpoints require a valid Bearer JWT header.

Method & Endpoint	Auth Required	Description & Payload
POST /api/auth/register/	Public	Register citizen account: { username, email, password, phone }.
POST /api/auth/login/	Public	Submit credentials. Returns OTP_REQUIRED with challenge ID.
POST /api/auth/verify-otp/	Pre-Auth	Validate 6-digit MFA OTP: { challenge_id, code }. Returns JWT tokens.
GET, POST /api/elections/	Authenticated	List user elections or create new draft election.
GET /api/elections/{id}/	Authenticated	Fetch election details, rules, and candidate roster.
POST /api/elections/{id}/voters/bulk-upload/	Creator / Admin	Upload CSV roster of eligible voter emails.
POST /api/elections/{id}/start/	Creator / Admin	Transition election status from scheduled to active.
GET /api/voter/elections/{id}/eligibility/	Voter	Check citizen eligibility and completed verification gates.
POST /api/voting/{id}/submit/	Verified Voter	Submit encrypted ballot. Returns cryptographic receipt hash.
GET /api/results/{id}/	Authenticated	View official tally breakdown and turnout statistics.
GET /api/ai/health/	Public	Health check for Gemini and Pinecone RAG configuration.
POST /api/ai/chat/	Authenticated	AI Assistant query: { message, election_id? }. Returns answer and sources.

5. Official DigiVote Knowledge Q&A; Reference

This section serves as the definitive reference manual indexed by the DigiVote AI Assistant. The questions and authoritative answers below provide exact procedural, legal, and operational guidance.

Q1: What is DigiVote?

DigiVote is an enterprise-grade digital voting system designed for secure, transparent, and legally sound elections. It combines strict multi-factor voter verification (email OTP, device security, and optional biometric face matching) with zero-knowledge encrypted ballot boxes. The platform guarantees voter privacy, prevents double-voting through database row locking, and provides cryptographic receipt tokens for independent verification.

Q2: How do I register as a citizen on DigiVote?

To register, navigate to the Register page, enter your full legal name, national or institutional email address, phone number, and a strong password. A confirmation email with a cryptographic activation link will be sent to your inbox. Click the link to verify your email. Once verified, you can sign in and access the Citizen Voter Portal.

Q3: How does multi-factor authentication (MFA) work during login?

When signing in, enter your email and password. If your credentials are correct, DigiVote generates a secure 6-digit One-Time Password (OTP) and dispatches it to your registered email address. Enter the OTP code within 5 minutes to complete authentication. You can also check 'Remember this device' to simplify subsequent logins on trusted personal computers.

Q4: How do I know whether I am eligible to vote in an election?

Eligibility is determined by the official voter roll configured by the election organizer. When you log in, go to 'Available Elections'. The system automatically compares your registered email with the election's eligible voter roster. Elections where you are eligible display an 'Eligible Voter' green badge. You can also ask the DigiVote Assistant: 'Am I eligible to vote in this election?' while viewing an election to receive an instant, personalized eligibility confirmation.

Q5: How does voter verification work before voting?

Before voting, you must complete the verification requirements established for that election. Depending on the organizer's configuration, this may include: (1) An email OTP challenge, (2) A webcam liveness scan to verify physical presence, or (3) Biometric facial matching comparing your live webcam photo to your registered profile photo using the ArcFace neural pipeline. Once verified, you are authorized to enter the Voting Booth.

Q6: How do I cast my vote in the Voting Booth?

Once identity verification is complete, enter the Voting Booth for your active election. Review the list of official candidates, their affiliations, and biographies. Select your chosen candidate and click 'Confirm & Submit Ballot'. A modal will request your final confirmation. Once submitted, the system records your anonymous ballot and issues an immutable cryptographic SHA-256 voting receipt.

Q7: Can anyone, including administrators or organizers, see who I voted for?

No. DigiVote enforces absolute ballot secrecy through its decoupled zero-knowledge architecture. When your ballot is cast, it is encrypted with AES-256-GCM and stored in the ballot box without any user ID, username, IP address, or timestamp linkage. Your receipt proves that a ballot was cast from your account, but it contains no data indicating which candidate you selected. Even system administrators with direct database access cannot correlate voters to their choices.

Q8: What happens if someone tries to vote twice?

Double voting is mathematically and architecturally prevented by DigiVote. When a vote submission begins, the backend opens an atomic transaction with row-level locking on your voter receipt record. If a voter receipt already exists or if multiple concurrent requests are fired simultaneously, the PostgreSQL unique database constraint on (voter_id, election_id) triggers a collision, rolls back the transaction, and returns an HTTP 400 'Double Voting Detected' security rejection.

Q9: Can I change my vote after submitting?

By default, votes are final and immutable once submitted to preserve election finality. However, if an election organizer explicitly enables the 'Allow Vote Change' rule in the election settings before polling opens, voters may update their vote while the election remains Active. When vote changing is disabled, a cast ballot cannot be modified or replaced under any circumstance.

Q10: What is the Voting Receipt and how do I use it?

A Voting Receipt is an SHA-256 cryptographic hash provided upon ballot submission (e.g., 7A3F9C1E...B402). It is uniquely derived from your voter identifier, the election ID, and a high-entropy salt. You can save or print this receipt to verify that your ballot was included in the official tally audit logs after the election closes without compromising ballot secrecy.

Q11: How do election organizers create and manage an election?

Organizers navigate to 'My Elections' and click 'Create Election'. They enter the title, description, and election category. Next, they configure the candidate slate (uploading bios and official photos), upload the eligible voter roll via CSV bulk ingestion, and establish verification gates (Email OTP, Webcam, Biometrics) and results visibility schedules. Once configured, the election is scheduled and transitions automatically to Active at the designated start time.

Q12: How are election results published and verified?

When an election reaches its scheduled end time or when an organizer manually completes polling, the election transitions to Completed. The tallying engine automatically aggregates the anonymous ballots. Depending on the election rules, results are displayed immediately, scheduled for a future ceremony, or manually published by the administrator. Certified reports with audit log signatures can be exported as PDF and CSV documents.

Q13: Why does the DigiVote Assistant refuse to recommend candidates?

The DigiVote AI Assistant is bound by strict civic neutrality guardrails. It is programmed to assist citizens with voting procedures, eligibility verification, election schedules, and technical support. It will never express an opinion, rank candidates, recommend who to vote for, or participate in political persuasion. If asked who to vote for, it will neutrally direct you to review candidate biographies in the Voting Booth.

Q14: Can the AI Assistant cast a vote on my behalf?

No. The AI Assistant has no voting capabilities or execution authority. It cannot cast ballots, confirm votes, change election configurations, or modify voter records. All voting actions must be performed directly and intentionally by the verified citizen inside the Voting Booth.